A Simple LP-Based Approximation Algorithm for the Matching Augmentation Problem

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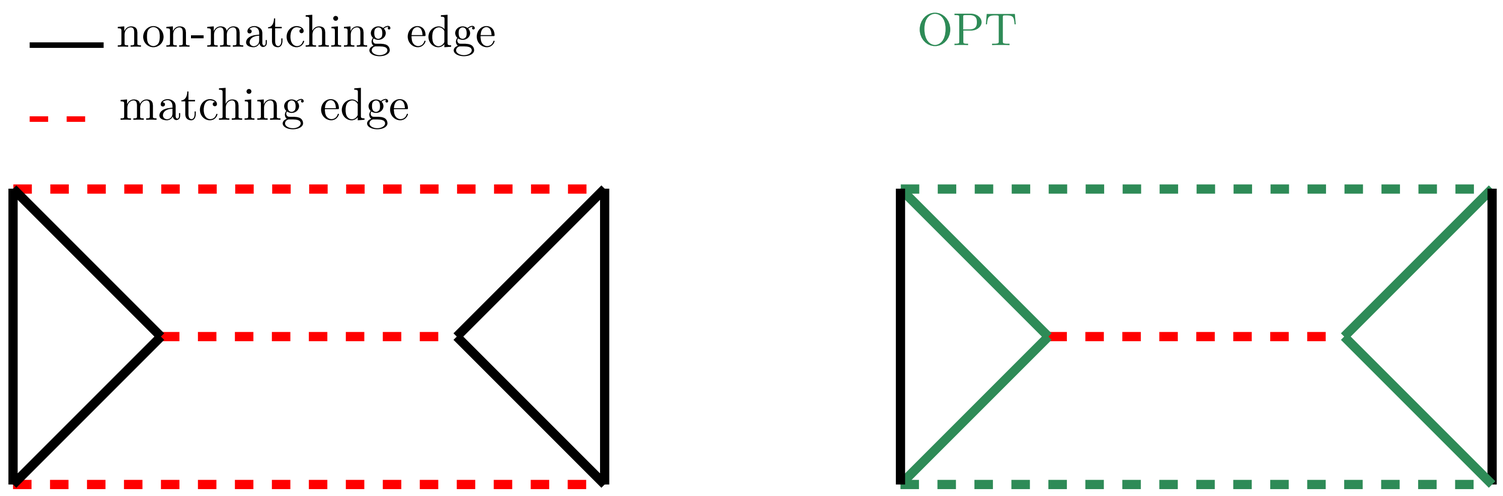
Matching Augmentation Problem (MAP)

- Input: A Graph, $G = (V, E)$ and a matching $M \subseteq E$
- Output : A 2-Edge Connected Spanning Subgraph $H = (V, E')$ minimizing $|E' \setminus M|$


non-matching edge
matching edge
OPT

Background

- MAP is a simplification of the more general weighted 2-ECSS, for which the best known polynomial-time algorithms give 2-approximations (Iterative Rounding, Primal Dual, Matroid Intersection)
- State of the Art for MAP: $5/3$-approximation [CDGKN20, CCDZ20]

Background Continued

- Tree Augmentation Problem (TAP) [N03, EFKN09, CKKK08, KN15, KN18, CG18, GKZ18, CCGK20, CTZ21, N21]
  - State of the Art: 1.393-approximation
  - Integrality Gap: ≈ 1.87
- Weighted Tree Augmentation Problem (WTAP)
  - Bounded Costs [A18, FGKS18]
  - General Case [FJ81, KT92, TZ21, TZ22]
    - State of the Art: 1.5-approximation
- FAP [AGT22]
  - State of the Art: < 2 -approximation

Our Results

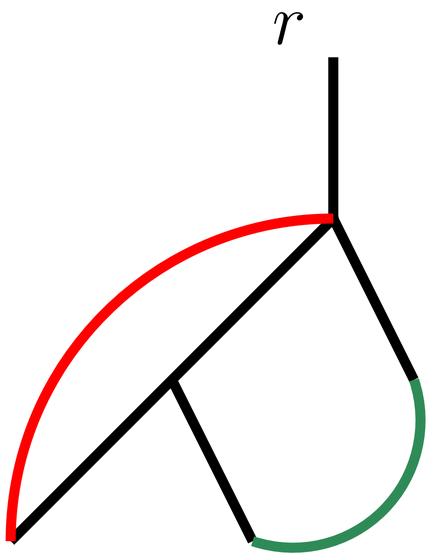
- Show that the natural LP formulation called the “cut LP” for MAP has integrality gap bounded below 2, using a simple algorithm
- Give improved lower bound of $4/3$ on the integrality gap of the cut LP

CUT LP

$$
\begin{aligned}
LP(G,F): \quad & \min \sum_{e \in E \setminus F} x_e \\
& \sum_{e \in \delta(S)} x_e \geq 2, \quad \text{for all } S, \emptyset \subsetneq S \subsetneq V \\
& 0 \leq x_e \leq 1, \qquad \forall e \in E.
\end{aligned}
$$

Helpful Tool

- The cut LP for instances of TAP containing only “up-links” is integral



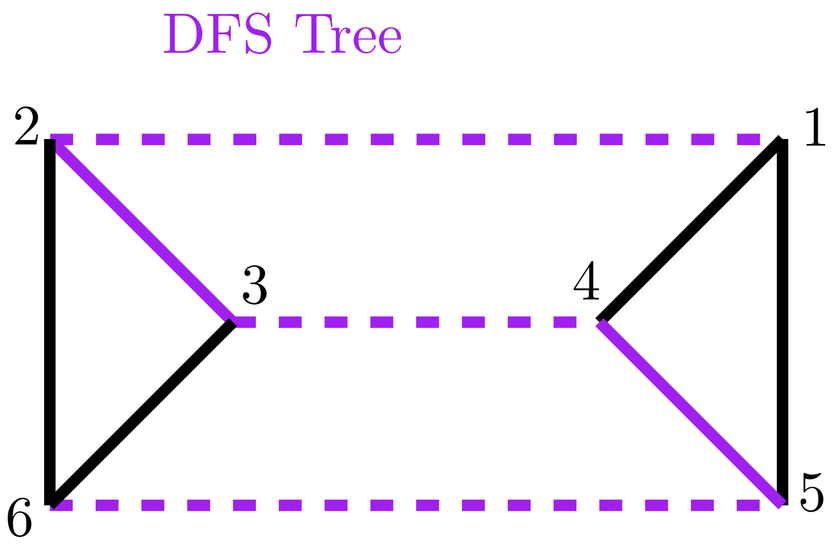
The Algorithm

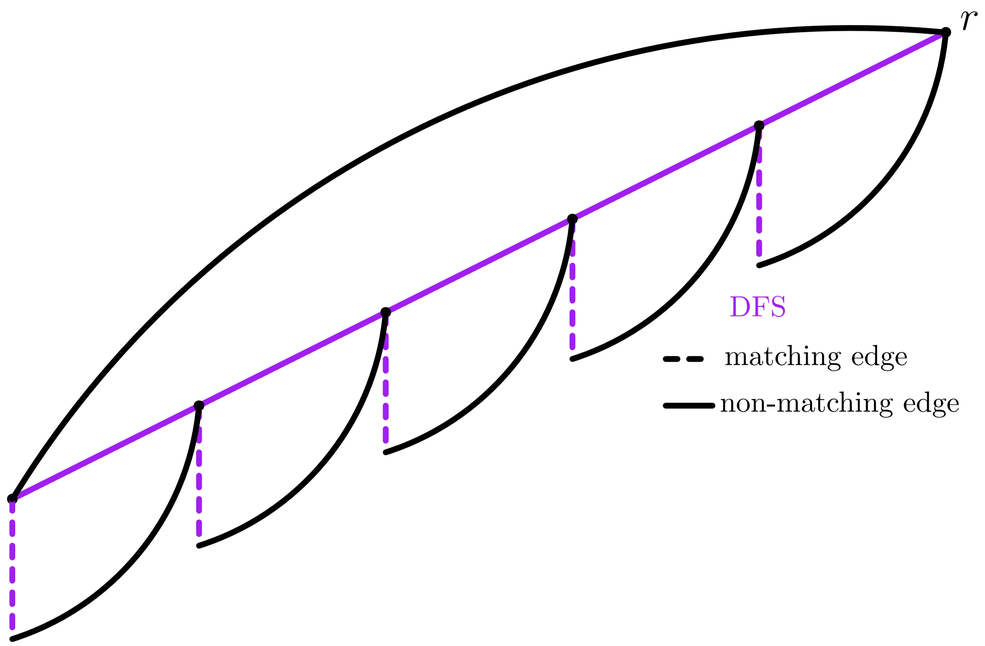
1. Compute an optimal extreme point solution x^* to $LP(G, M)$.

2. Let $E' = \{e \in E, x_e^* > 0\}$be the support of x^*, and run a DFS on the support graph $G' = (V, E')$ which always give priority first to an available light edge and second to the available heavy edge e maximizing x_e^*.

3. Compute an optimum augmentation A to the TAP problem with respect to the DFS tree T computed in the previous step and return $H = T \cup A$.

Examples





Warm Up: 2-approximation

- Well known that the cut LP for TAP where the input is a DFS is integral
- Cost of Final Solution : $c(T) + c_T(x)$, where T is our DFS tree, and c_T is the cost for the resulting TAP instance, and x is an optimal solution for the TAP cut LP
- Gives us a 2-approximation as $c(T) \leq c(x^*)$ and $c_T(x) \leq c(x^*)$

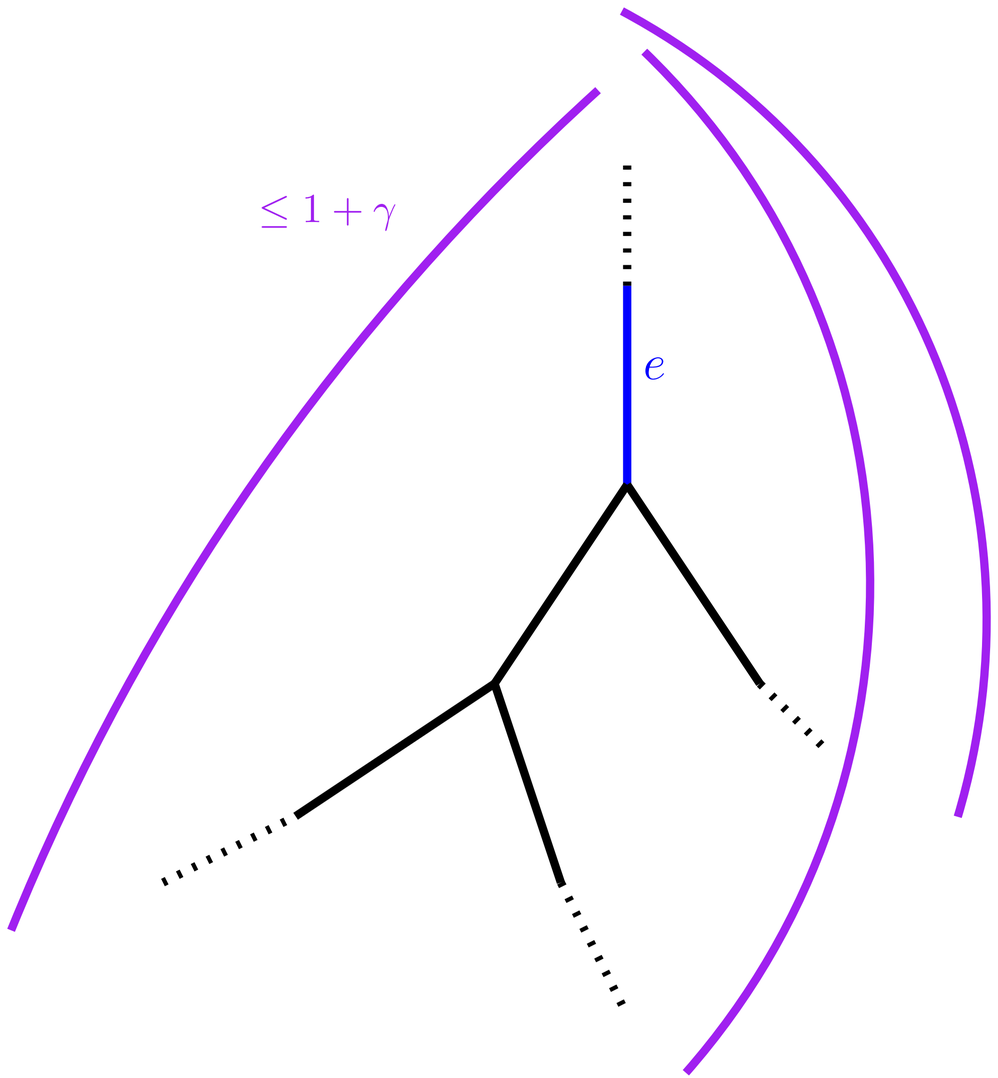
Warm Up: x^* is Half-Integral

- Note that $c(x^*) \geq n - |M|$
- $c(T) = n - |M|$
- $c_T(x) \leq c(x^*) - \dfrac{n - |M|}{2}$
- Obtain $3/2$-approximation

Strategies to Break 2 in General

- Show that either (i) $c(T) < (1-\epsilon)c(x^*)$ or (ii) $c_T(x) < (1-\epsilon)c(x^*)$
- Easy Case: $c(x^*) \geq (1+\epsilon)(n - |M|)$, and so we assume $c(x^*) = n - |M|$ and prove (ii)
- Note that $c_T(x) \leq c(x^*_{E \setminus T}) = c(x^*) - c(x^*_T)$
- To prove (ii) we show that either a) We can scale down $x^*_{E \setminus T}$ to get a feasible TAP solution, or b) $c(x^*_T) > \epsilon(n - |M|)$.

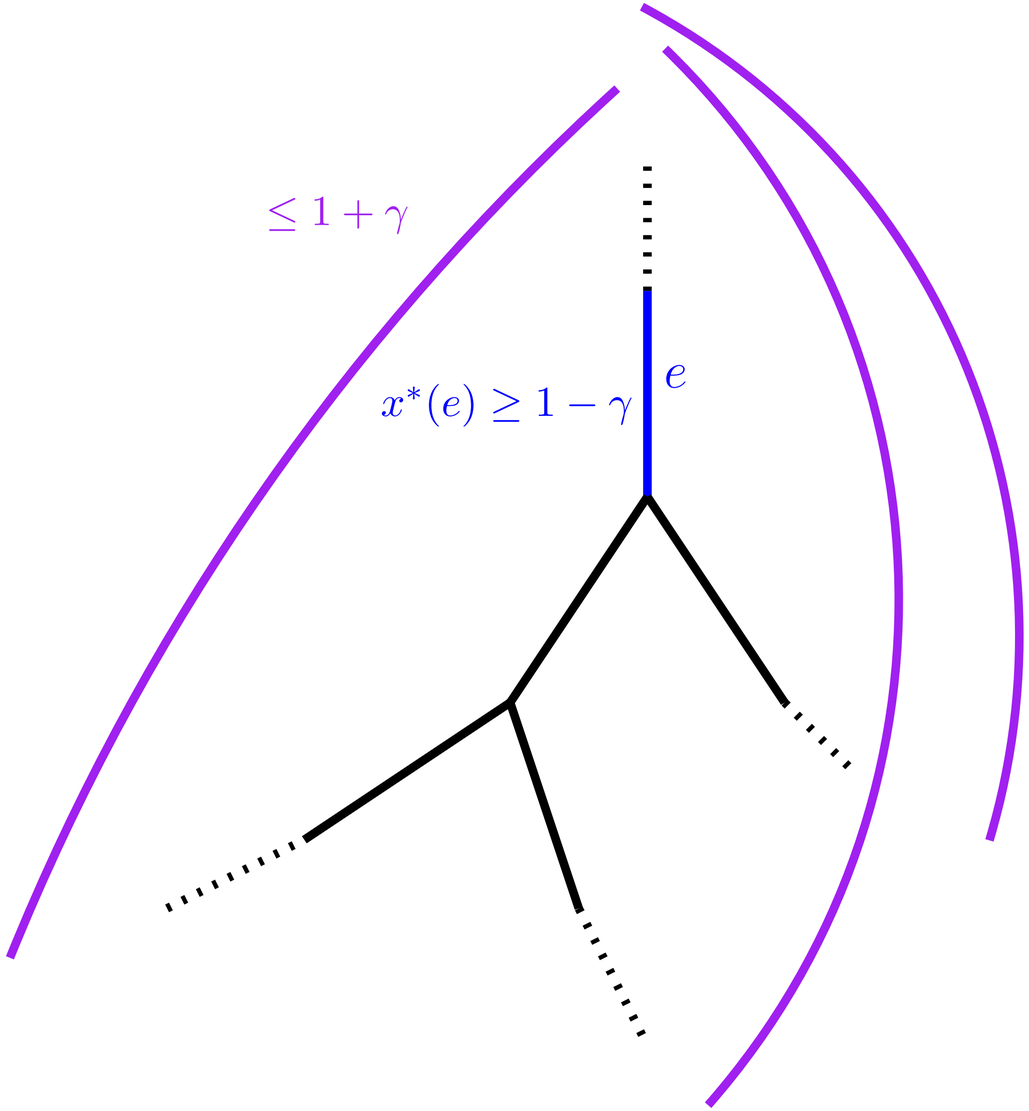
γ-tightness



Case 1: Few γ-tight cuts, t

- Then the total value on back-edges is at most $(1+\gamma)t$
- Can perform scaling on back-edges in x^* to get an improved solution to tree augmentation problem

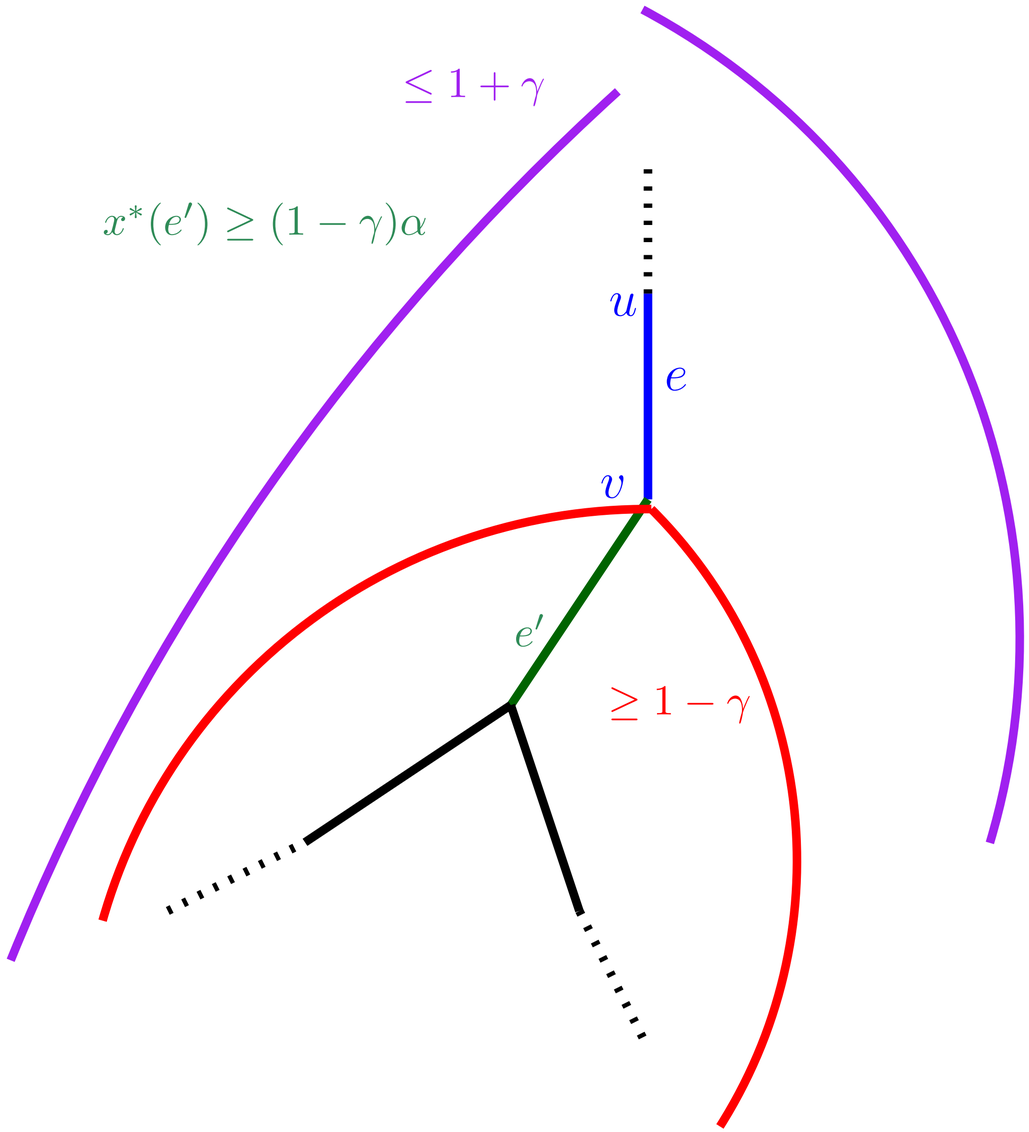
Case 2: Lots of γ-tight non-matching cuts



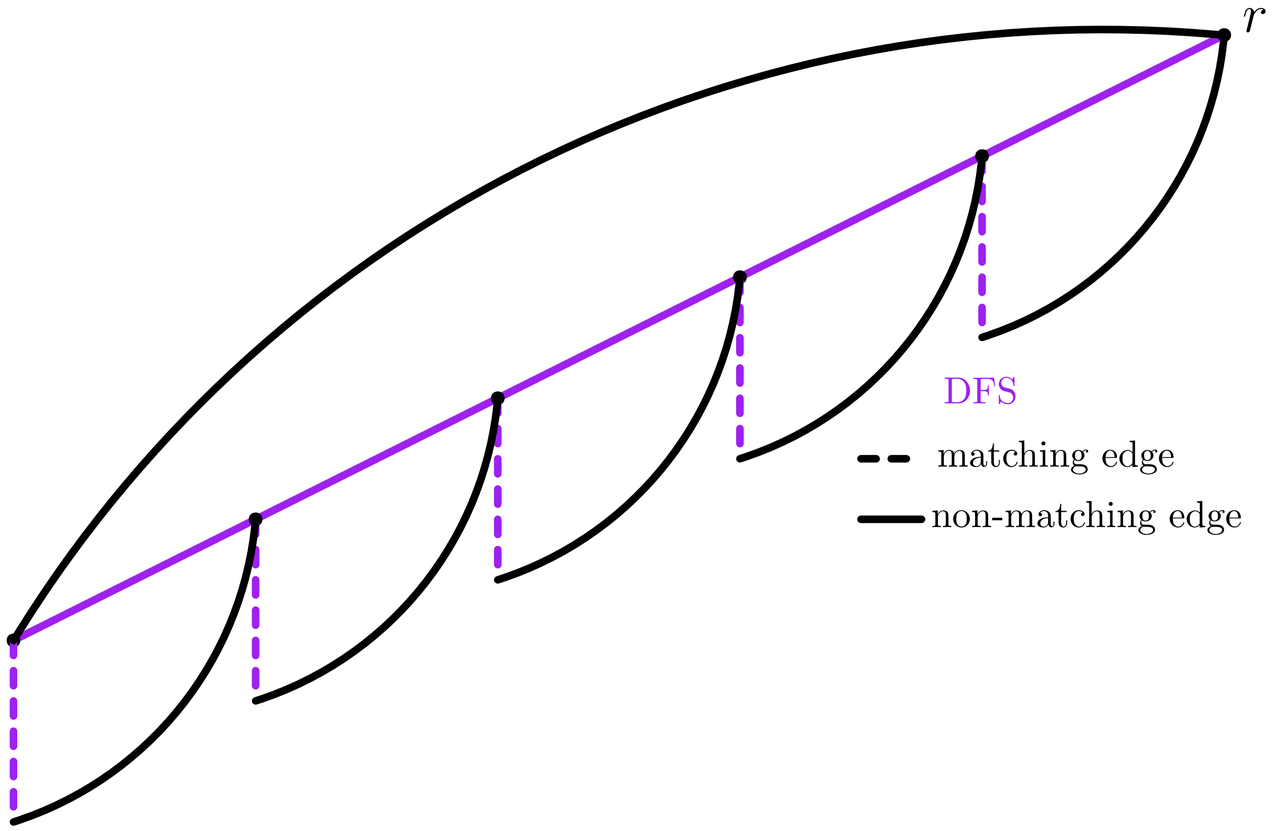
Case 3: Lots of γ-tight matching cuts, $e = uv$, where v is not a leaf of T

- Definition: We say a vertex v is α-fractional it is has more than $\frac{1}{\alpha}$ incident fractional edges
- Lemma: If x^* is an extreme point solution of the cut LP, then there are at most $2n - 1$fractional edges in G. Moreover, for any $\alpha > 0$, there are at most $4\alpha n$ α-fractional vertices with respect to x^*.

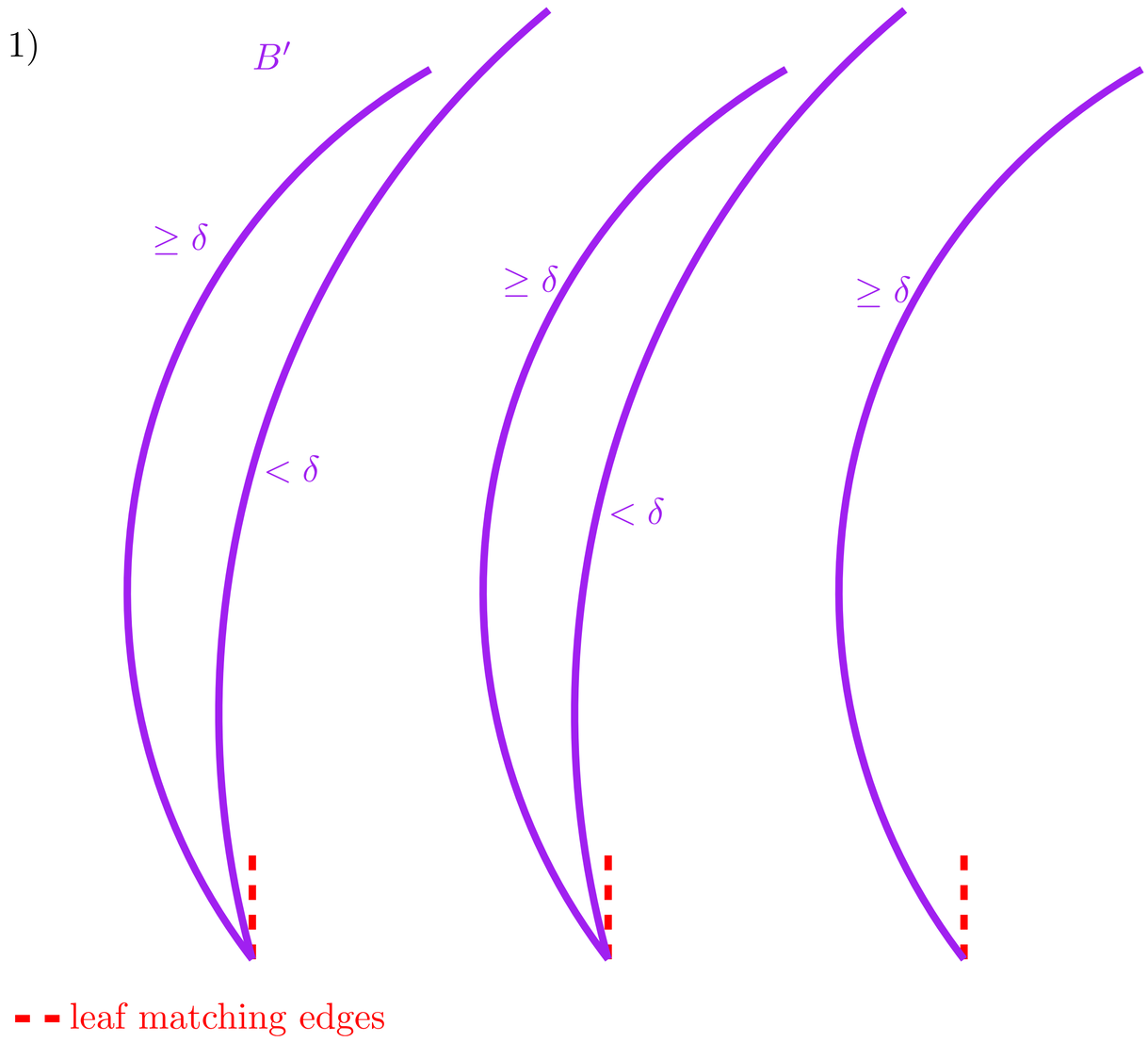
Case 3: Lots of γ-tight matching cuts, $e = uv$, where v is not a leaf of T, continued



Return to Bad Example



Case 4: Lots of γ-tight matching cuts, $e = uv$, where v is a leaf of T, and few non-leaf γ-tight matching cuts



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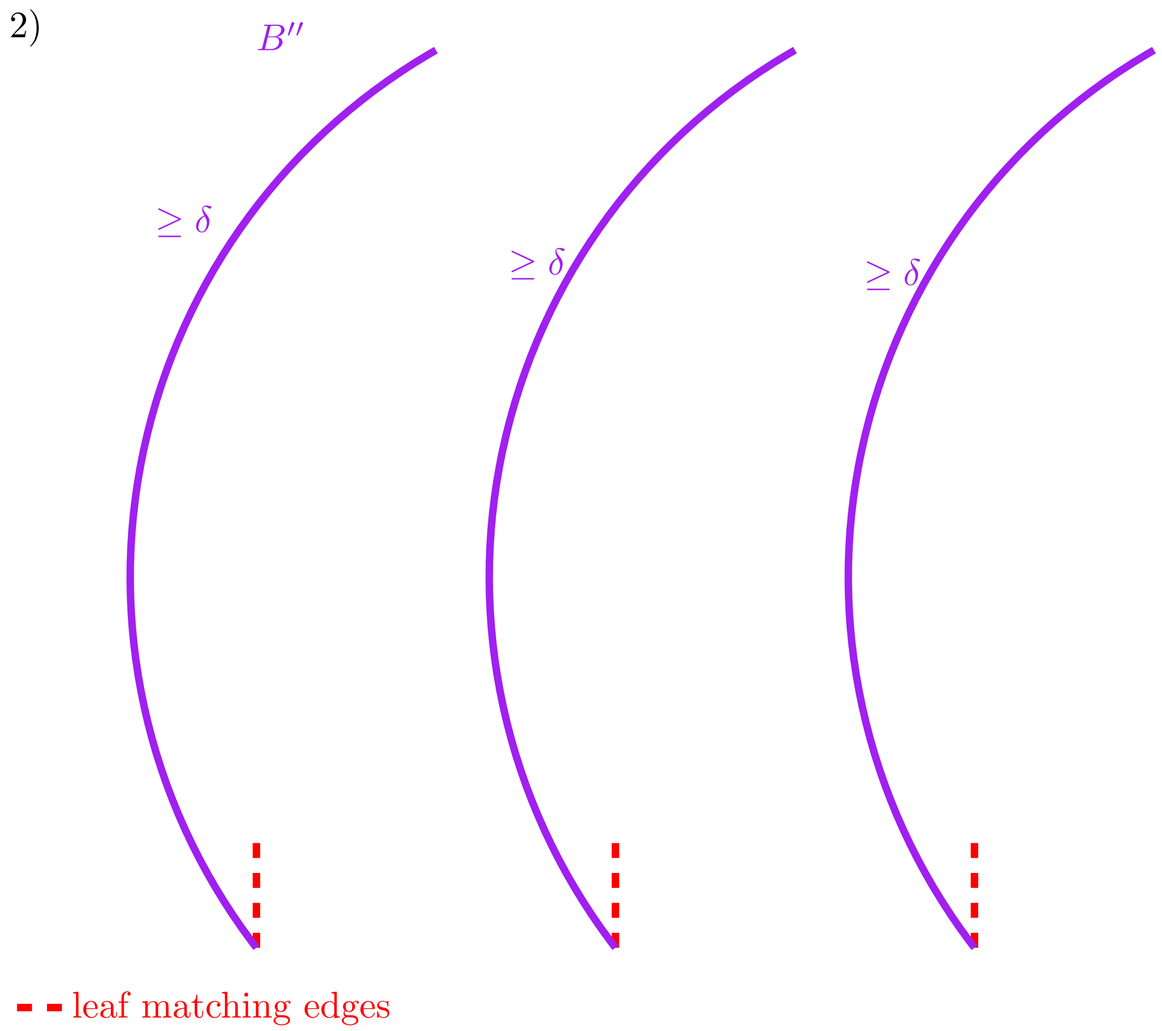
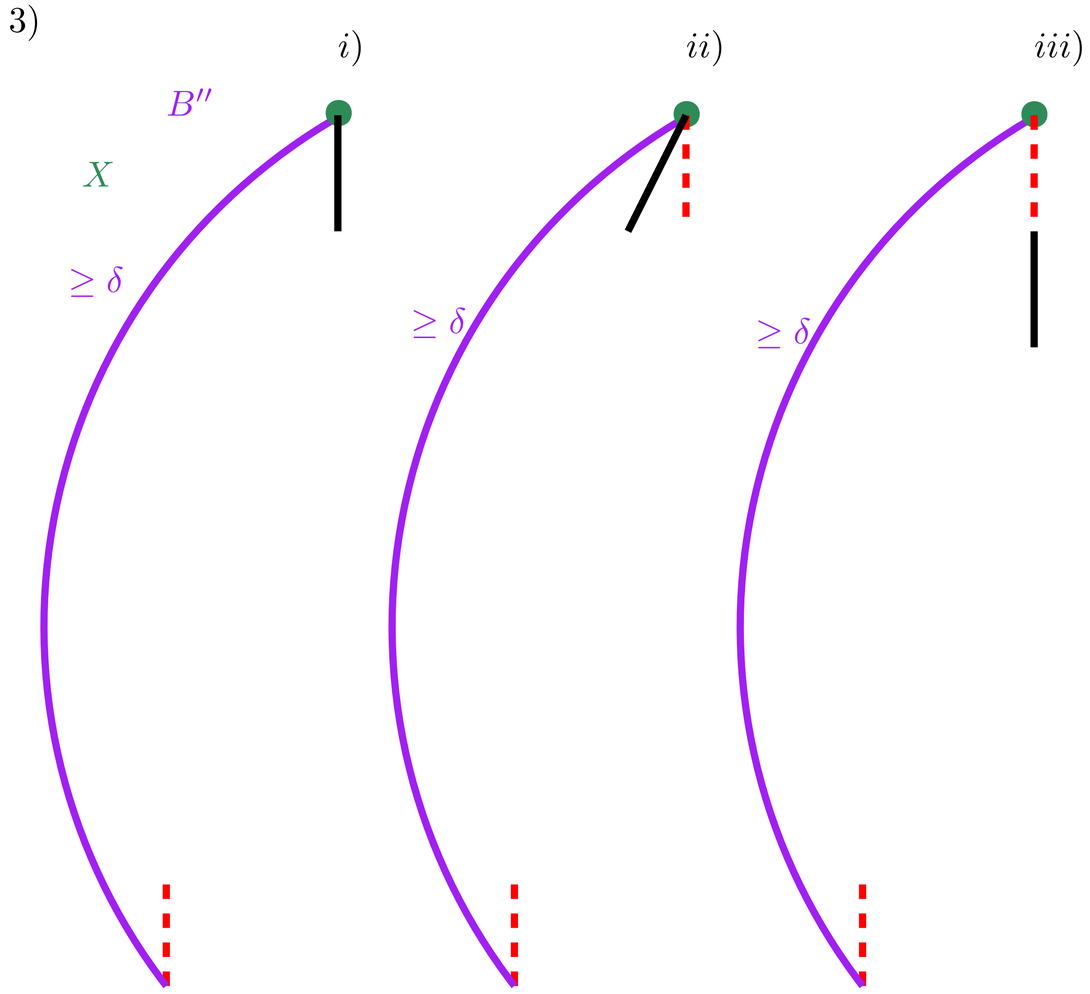


Case 4: Lots of γ-tight matching cuts, $e = uv$, where v is a leaf of T, and few non-leaf γ-tight matching cuts, continued



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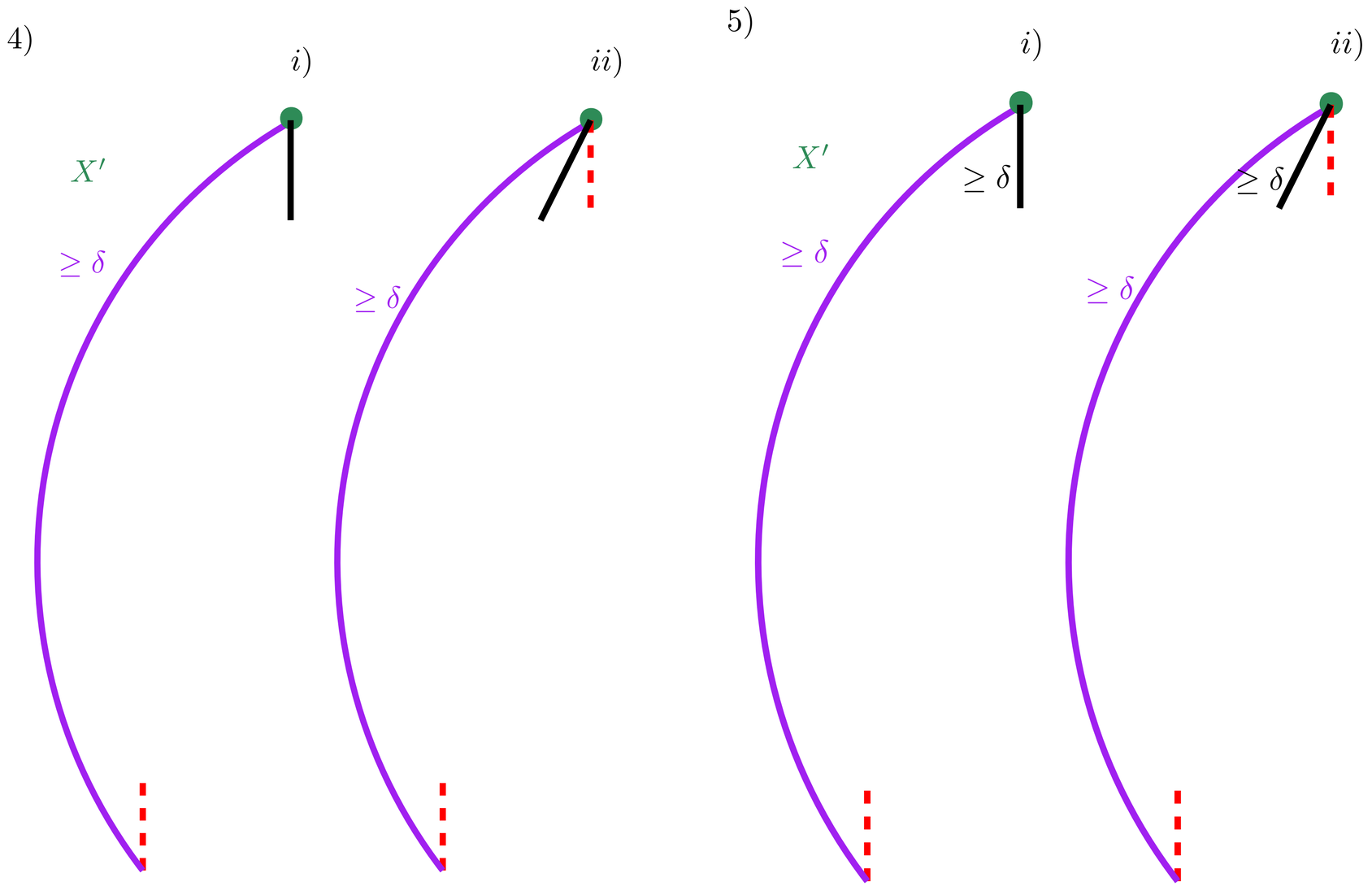
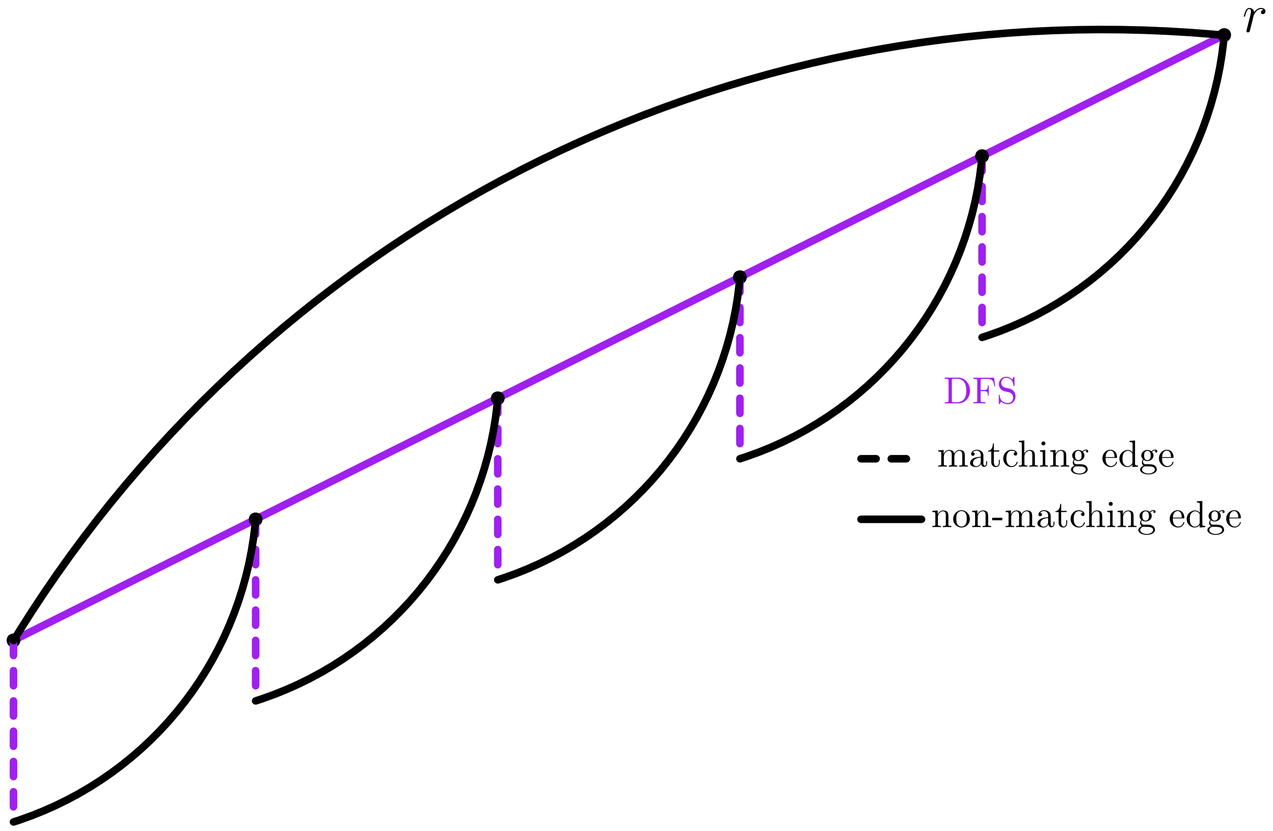


Case 4: Lots of γ-tight matching cuts, $e = uv$, where v is a leaf of T, and few non-leaf γ-tight matching cuts, continued

- $c(T) \geq \delta |X'|$, but $|X'|$ must be proportional to the number of γ-tight leaf edges, for otherwise since we are not in Case 3, we would obtain that $c(x^*) >> n - |M|$

Return Again to Bad Example



Open Problems

- Giving a better than 2 -approximation for 2-ECSS in the weighted setting
- Improving this algorithm and its analysis, extension of techniques for related problems
- Cut LP integrality gap below 2 for TAP [N21], what is the integrality gap of the cut LP for FAP instances?